claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a5c4ad7e429a7e7aa.jsonl`
- SHA-256: `6c7a5a48250b697cd611423656d441e924223133f5fcbb54add3673133859efa`
- Source modified: `2026-06-06T17:31:12+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:14.123Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/html/2505.03991v1
Title: Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges (arXiv 2025)
Found via: broad/primary search

Task
1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:19.834Z)

I'll fetch the source and extract key claims relevant to the research question.

3. user (2026-06-06T17:30:48.744Z)

Claims on Sports Event Detection in Racket Sports

Publication Details
- **Type**: Peer-reviewed survey (arXiv:2505.03991v1)
- **Date**: May 6, 2025
- **Authors**: Xu et al. (Deakin University, Agriculture Victoria Research, Paralympics Australia)

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(1) Method Families for Action Spotting / Precise Event Spotting

Temporal Action Localization (TAL):
- "Global-to-Local methods generate action proposals through sliding windows or pre-defined anchor structures"
- "Local-to-Global methods independently evaluate each frame to predict the likelihood of action start, action end"

Action Spotting (AS):
- "AS proposes an alternative to TAL by representing each action with a single keyframe rather than a temporal interval"
- Methods include: 3D CNNs, 1D convolutions, Transformers, multi-scale architectures, regression-masking approaches

Precise Event Spotting (PES):
- "PES requires extremely strict detection thresholds, typically constrained within 1?5 frames (approximately 33?200 milliseconds)"
- Recent approaches use "Vision-language models" and "Temporal Discriminability Enhancer Encoder-Decoder (T-DEED)"

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(2) Datasets for Racket Sports

Tennis

1. **Tenniset** (2017): "Five full-match videos from the 2012 London Olympic Games...six event types, such as 'set,' 'hit,' and 'serve'"
   - Annotation: Temporal intervals
   - License/Availability: From YouTube sources

2. **Tennis Dataset** (2022, Hong et al.): "3,345 video clips from 28 professional tennis matches...six classes, including 'player serve ball contact,' 'regular swing ball contact,' and 'ball bounce'"
   - Annotation: Frame-level spotting
   - Frame rate: 25 or 30 FPS

Table Tennis
1. **OpenTTGames** (2020): "12 high-definition table tennis match videos...4,271 labeled events...three event types: ball bounces, net hits, and empty events"
   - Frame rate: 120 FPS
   - Annotation: Frame-level precision

2. **P2A** (2024): "2,721 untrimmed broadcast videos with a total length of 272 hours...14 fine-grained stroke classes, further grouped into 8 higher-level action categories"
   - Sources: "Olympics, World Cups, and World Championships"
   - Validation: "Professional validation by players and referees"

Badminton
1. **Badminton Olympic** (2018): "27 badminton match videos...12 action types, such as 'serve,' 'backhand,' and 'smash'"
   - Annotation: Time-interval format

2. **BadmintonTrack** (reference [73]): "77,000 annotated frames from 26 unique singles matches"
   - Camera view: "Overhead broadcast-view camera"

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(3) Reported Accuracy/Metrics for Specific Methods

Soccer (SoccerNet) Benchmarks

Method	Year	Metric	Score	Notes
Giancola et al.	2018	Loose mAP	31.37%	First baseline
RMS-Net	2021	Loose mAP	63.49%	Regression & masking
Zhou (Transfer Learning)	2021	Tight mAP	47.05%	Soccer-specific features
STE	2022	Tight mAP	58.29%	"1D convolutions and fully connected layers"
SpotFormer	2022	Tight mAP	60.90%	Multi-feature Transformers
Soares et al.	2022	Tight mAP	65.07%	Dense detection anchor
ASTRA	2023	Loose mAP	79.21%	"Transformer encoder-decoder" with audio
COMEDIAN	2024	Tight mAP	73.10%	"Self-supervised learning & knowledge distillation"

Tolerance Windows:
- **AS (Action Spotting)**: "5 to 60 seconds" or "5?60 frames"
- **PES (Precise Event Spotting)**: "0?2 frames" or "1?5 frames (approximately 33?200 milliseconds)"

Tennis/Fine-Grained Evaluation
- "Frame-level precision is critical in table tennis, where even minor inaccuracies can significantly impact the determination of ball bounce locations"

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(4) Compute Requirements

Explicitly mentioned:
- SpotFormer: "The use of multiple feature extractors significantly increases computational cost and inference time, making it less practical for real-time or resource-constrained applications"

Comparison noted:
- STE achieves "strong performance...while maintaining strong performance" via "1D convolutions" versus 3D CNNs (which are "known for their high computational cost")

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(5) Audio/Multimodal Event Detection

Audio fusion methods:
1. **Vanderplaetsen et al.** (2020): "Different fusion strategies...found that late fusion?integrating audio and visual features just before the fully connected layer?yielded the best performance"

2. **ASTRA** (2023): "Transformer-based encoder-decoder model...processes both visual and audio embeddings using learnable queries"

Proposed future directions:
- "Attention-based cross-modal transformers and gated attention mechanisms, capable of dynamically weighting and integrating audio and visual modalities"
- "Exploiting commentary audio through automatic speech recognition and natural language processing"

Vision-language fusion (Tran et al., 2024):
- "Integrates GLIP...a vision-language model that processes a single image and a textual prompt describing the target objects to detect (e.g., 'ball,' 'referee,' 'card')"
- "First use of vision-language models in PES"

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(6) Self-Supervised / Weakly-Supervised / Distillation

Active Learning:
- Giancola et al. (2023): "First active learning framework for AS...predictions ranked using uncertainty and entropy measures to identify the most informative samples"
- Achievement: "Results comparable to state-of-the-art methods using only one-third of the dataset"

Self-Supervised Learning & Knowledge Distillation:
- **COMEDIAN** (Denize et al., 2024): "First AS-specific framework that combines self-supervised learning (SSL) with knowledge distillation (KD)...spatial transformer trained using Momentum Contrast (MoCo)"
- "Both components further enhanced using a Soft Contrastive loss (SCE)...distills knowledge from a bank of rich spatiotemporal features"

Proposed future work:
- "Development of self-supervised frameworks that leverage temporal consistency, contrastive learning, and multimodal alignment to extract meaningful event representations from unlabeled or weakly labeled video data"

Key Gaps Identified by Authors

- **Generalization**: "Most existing AS and PES models...perform well within their target sports but often exhibit limited capability for generalization"
- **Annotation costs**: "Creating extensive labeled datasets for sports event detection is costly, labor-intensive, and often demands expert domain knowledge"
- **Multimodal limitations**: "Existing AS and PES methodologies predominantly rely on visual data...straightforward fusion methods...do not fully capture the complex interactions between modalities"

4. assistant (2026-06-06T17:30:52.465Z)

This is a relevant, high-quality survey. Let me extract the most falsifiable claims bearing on the research question and submit structured output.

5. user (2026-06-06T17:31:01.929Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:11.882Z)

I fetched and extracted from the arXiv survey "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges" (arXiv:2505.03991v1, May 6, 2025; Deakin University et al.). It is a primary academic survey, directly on-topic for the research question.

Five falsifiable claims submitted via StructuredOutput, three rated central:
- PES vs AS tolerance gap (1-5 frames / ~33-200 ms for precise event spotting) ? defines the difficulty regime for frame-accurate rally boundaries.
- Audio-visual fusion is competitive/complementary (late fusion best; audio-augmented ASTRA = 79.21% loose mAP on SoccerNet) ? supports the user's interest in local audio-based hit detection.
- SSL + knowledge distillation is a proven top approach (COMEDIAN = 73.10% tight mAP) ? validates the teacher->student distillation direction.

Two supporting claims: active learning matching SOTA with 1/3 the labels (relevant to their n=2 data-starvation problem), and badminton dataset characteristics (Badminton Olympic, BadmintonTrack ? stroke-typed/overhead, not rally-boundary labeled).

Note on a gap relevant to their pipeline: the survey covers datasets but the extraction did not surface explicit license terms for the racket-sport datasets, and none of the badminton sets described are rally-start/end segmentation labeled (they are stroke/action-type spotting), so commercial-license verification would need to go to the dataset sources directly.